Stochastic Risk Evaluation Scheme for Wave-off Decision Making in Automatic Aircraft Carrier Landing

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Abstract

One of the greatest challenges for the air operations of fully automated aircraft carrier is to design an autonomous decision making system. In this paper, a probability-based risk evaluation scheme is proposed to make a wave-off, which is equivalent to go-around in ground landing, and the wave-off decision plays an important role in the terminal portion of landing. By the proposed risk evaluation scheme, the most probable consequence of the landing attempt can be forecasted. The risk evaluation scheme introduces six variables which are related to a glide path error, and they are assumed to follow the Gaussian probability distribution. The Monte Carlo simulation results with X-Plane flight simulator show that there exists a significant positive correlation between the estimated probabilities and the actual occurrence rate.

Keywords: aircraft carrier landing, stochastic risk evaluation, decision making, wave-off decision, software-in-the-loop simulation (SILS)

Introduction

In comparison with ground landing, aircraft carrier landing has many risk factors: roll, pitch, and heave motions of a carrier, angled and short flight deck for landing, small ramp-to-hook clearance and a steep vertical angle of a glide path. Above all, a moving aircraft carrier creates a turbulent flow around the control tower and behind the carrier, which may lead to a significant degradation of landing control performance. Considering hard conditions such as low approach speed, limited space, and a very limited amount of time available [1, 2], it is required to evaluate the risk appropriately during the final stage of landing, which will help to make a timely and adequate decision on whether to abort landing and enter into a wave-off pattern or not.

Landing signal officers (LSOs) have taken responsibilities for the wave-off decision in manual landing, and therefore, various decision support tools have been developed to assist the landing [3, 4]. In automatic aircraft carrier landing, most of wave-off decision making techniques mainly consider a ramp-to-hook clearance, because an insufficient clearance could result in a ramp strike. Johnstone [5] defined the potential carrier ramp strike wave-off criteria by applying a military power wave-off trajectory resulting from a specific initial condition. In [6, 7], a back-propagation neural network was applied to the automatic landing problem to obtain functions which give a ramp-to-hook clearance for the entire region behind the carrier and a quantitative ramp strike risk. Durand et al. [8, 9] suggested that the operational performance indices can be determined from the terminal glide path error variance data.

In the near future, unmanned aerial vehicles will become an important part of aircraft carrier the air operations, which will introduce further complexity to the carrier operations [10]. Therefore, the autonomous wave-off decision system should be developed to replace human LSOs for the effective operations. In this study, a risk evaluation scheme for decision making in the final phase of the aircraft carrier landing is proposed. The scheme considers statistical characteristics of six key variables: vertical/horizontal glide path error, roll/pitch angle, sink rate, and sideway velocity (a velocity component perpendicular to the runway). All the variables are affected by wind, air wake disturbances, and normal air turbulence. The proposed risk evaluation scheme is basically an extension of method in [8, 9], and is designed to meet the requirements of fully autonomous aircraft carrier landing. In this study, ship motion is assumed to be small. To demonstrate the performance of the proposed risk evaluation scheme, the software-in-the-loop simulation (SILS) is carried out with X-Plane flight simulator.

Decision Making System

Aircraft Carrier Landing Procedure

In automatic carrier landing, an aircraft follows the prescribed landing procedure as described in Fig. 1. Passing through the radar acquisition window in Fig. 1, the approaching aircraft receives glide slope and localizer signals, and the aircraft performs glide path capture [11]. Combined with distance information, a vertical and a horizontal glide path errors can be made. As the approaching aircraft flies close to the carrier, the aircraft experiences more severe turbulent air wake of the carrier in addition to the increasing ground effect of sea surface. Typically, from 12.5 and 1.5 seconds before touchdown, except a manual wave-off case by the LSO, wave-off decision is automatically made considering the current degree of risk. The amount of the risk can be evaluated based on either wave-off boundaries in a conventional manner or risk evaluation scheme.

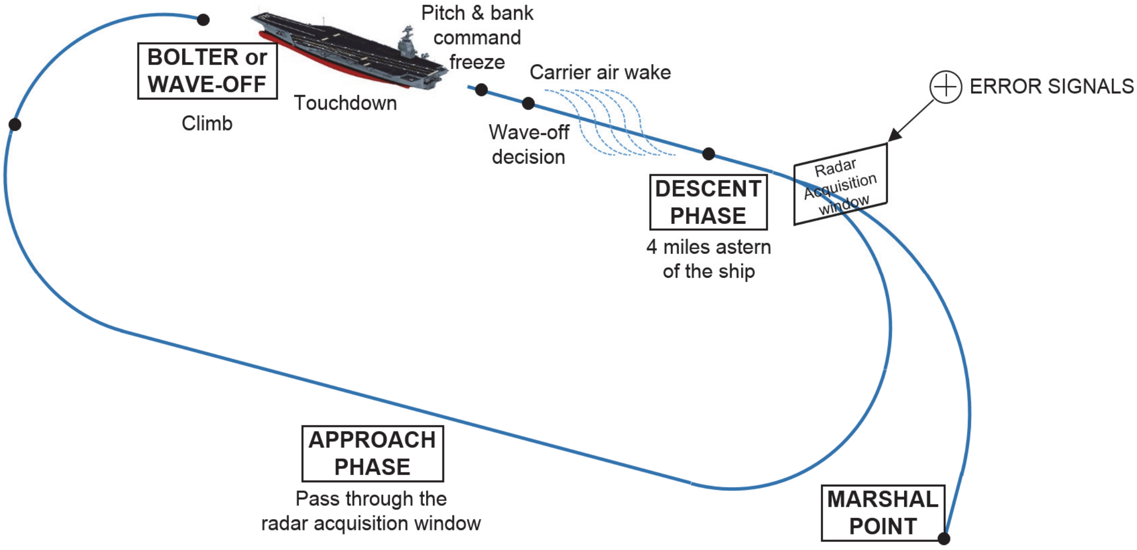

Fig. 1: The standard procedure of automatic aircraft carrier landing

The approaching aircraft continues tracking a glide path in the descent phase with the help of the guidance law of instrument landing system (ILS). The ideal glide path is independent of the carrier motion within the Fresnel lens optical landing system (FLOLS) stabilization limits (±6° pitch and ±10° roll) as shown in Fig. 2 [12]. The glide slope and localizer errors δ_v and δ_h are proportional to the deviation angle of the approaching aircraft from the ILS glide path within the proportional deflection area limit: usually ±0.7° in the vertical direction and ±3.0° in the horizontal direction. Let us define the vertical and horizontal glide path errors ε_v and ε_h as angular errors multiplied by the distance between the two objects.

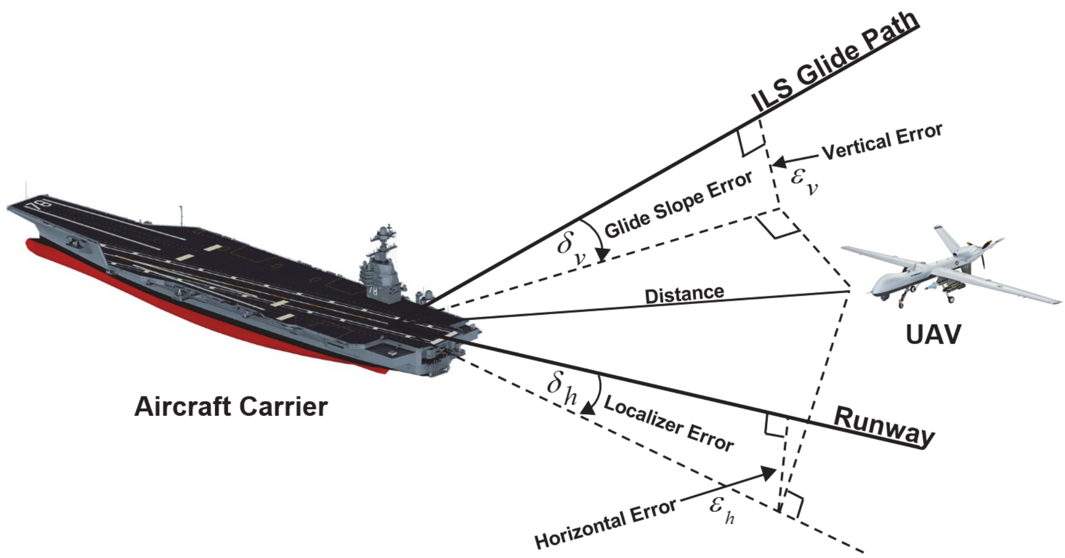

Fig. 2: Definitions of the error variables used in aircraft carrier landing

Six Key Variables

Six key variables are considered in the risk evaluation scheme: vertical error, horizontal error, roll angle, pitch angle, sink rate, and sideway velocity. These variables are directly related to the glide path tracking error, which can lead to a crash when they deviate excessively from the nominal values. In order for the approaching aircraft to land safely on the aircraft carrier, all the key variables should be remained within the allowable range in the terminal landing phase, especially at the moment of touchdown. Without wave-off maneuver, the results of landing attempts eventually fall into one of three cases depending on the values of the six variables. Table 1 summarizes all three cases of the results: land (landed securely on the carrier), bolter (take-off after unarrested touchdown), and fail (destructive crash). A symbol O indicates that the corresponding variable resides in the allowable range, while X indicates that it does not.

Table 1: Three possible results of a landing attempt

Result / Variable	Recover	Bolter	Fail
Vertical error ε_v	O	X	
Horizontal error ε_h	O	O	
Roll angle ϕ	O	O	When at least one of the factors deviates from the nominal value too much
Pitch angle θ	O	O	
Sink rate $\dot{h}$	O	O	
Sideway velocity v_s	O	O	

Risk Evaluation

According to Refs. [13-15], it is assumed that the six key variables have Gaussian distribution. This is because the motion of the aircraft near the carrier is perturbed primarily by the carrier air wake which has stochastic characteristics. The probabilities of three possible results can be calculated using the mean and variance of the variables during an appropriate time. Figure 3 illustrates six probability distributions according to the computed mean and variance value for about 40 seconds before touchdown. It can be regarded that the later data are more important than the earlier data, but they have almost equal importance because there is no flare maneuver at all. Therefore, the aircraft reacts to disturbances in the exactly same manner throughout the entire time interval. The undesirable range of the variables corresponding to bolter (blue) and fail (red) is also shown in Fig. 3, where the bound values can be adjusted for other types of aircraft.

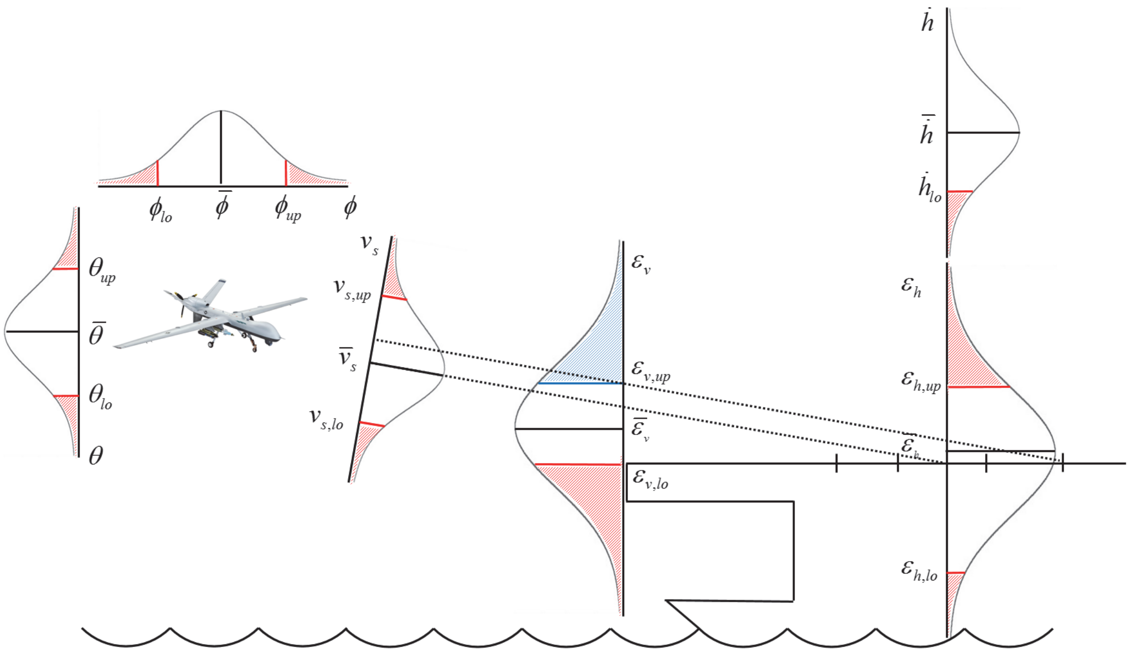

Fig. 3: Definitions of the probabilities used in the risk evaluation scheme.

For example, ramp strike risk can be computed as follow.

$$\begin{aligned} P_{\mathrm{RS}} &= \frac{1}{\sigma_v\sqrt{2\pi}}\int_{-\infty}^{e_{v,lo}} \exp\left[-\frac{\left(x-\bar{e}_v\right)^2}{2\sigma_v^2}\right]dx \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\frac{e_{v,lo}-\bar{e}_v}{\sigma_v}} \exp\left[-\frac{z^2}{2}\right]dz \\ &= \frac{1}{2}\left[1+erf\left(\frac{e_{v,lo}-\bar{e}_v}{\sigma_v\sqrt{2}}\right)\right] \end{aligned} \tag{1}$$

The other five types of risk can be computed in the same manner: P_{UL} , P_{OL} , P_{BA} , P_{HL} and P_{RB} . Table 2 summarizes the meaning of the symbols.

Table 2: The classification of the risk types and related variables

Type of risk	**Symbol**	**Related variable**	**Condition**
Ramp strike	P_{RS}	e_v	Excessive negative vertical error
Unarrested landing	P_{UL}	e_v	Excessive positive vertical error
Off-runway landing	P_{OL}	e_h	Excessive horizontal error
Bad attitude	P_{BA}	ϕ , θ	Bad attitude at the moment of the touchdown
Hard landing	P_{HL}	$\dot{h}$	Excessive sink rate
Runway breakaway	P_{RB}	v_s	Excessive sideway velocity

Under the assumption of mutual independence, the probabilities of the three possible outcomes are obtained as follows.

$$\begin{aligned} P_{\mathrm{recover}} &= \left(1-P_{\mathrm{RS}}\right)\left(1-P_{\mathrm{UL}}\right)\left(1-P_{\mathrm{OL}}\right)\left(1-P_{\mathrm{BA}}\right)\left(1-P_{\mathrm{HL}}\right)\left(1-P_{\mathrm{RB}}\right) \\ P_{\mathrm{bolter}} &= P_{\mathrm{UL}}\left(1-P_{\mathrm{RS}}\right)\left(1-P_{\mathrm{OL}}\right)\left(1-P_{\mathrm{BA}}\right)\left(1-P_{\mathrm{HL}}\right)\left(1-P_{\mathrm{RB}}\right) \\ P_{\mathrm{fail}} &= 1-\left(1-P_{\mathrm{RS}}\right)\left(1-P_{\mathrm{OL}}\right)\left(1-P_{\mathrm{BA}}\right)\left(1-P_{\mathrm{HL}}\right)\left(1-P_{\mathrm{RB}}\right) \end{aligned} \tag{2}$$

Now, two techniques can be applied to the wave-off decision. First, users set a threshold based on P_{fail} so that they allow possible bolters, because it is not fatal even though it may be dangerous to a certain extent. Second, users can set a threshold based on P_{recover} , which means that users may want to exclude or minimize possible bolters as well as a total crash, and prefer a wave-off to a bolter.

Numerical Simulation

Figure 4(a) shows the overall architecture of the simulator [16], and the proposed risk evaluation scheme is implemented in the existing decision making system as shown in Fig. 4(b). The decision making system runs throughout the whole landing process, and it assigns a proper operation mode to the aircraft. The guidance and control system designed in [17] is also adopted for autonomous carrier landing guidance and control. MQ-9 Reaper and USS Nimitz model of X-Plane, which are used in the simulation, are shown in Fig. 4(c). At the end of each simulation, the result is identified and classified into three classes: land, bolter and fail. In the numerical simulations, risk evaluation is performed for various turbulent conditions. For every turbulence level from 0.0 to 6.0 with an interval of 0.4, a thousand landing attempts are made. The X-Plane turbulence model is analyzed as shown in Fig. 5, and Table 3 summarize how the X-Plane turbulence level corresponds to Beaufort force and Pierson-Moskowitz sea state. Note that all aircraft carrier air operations in sea state above 6 are virtually prohibited even for manned aircraft in general.

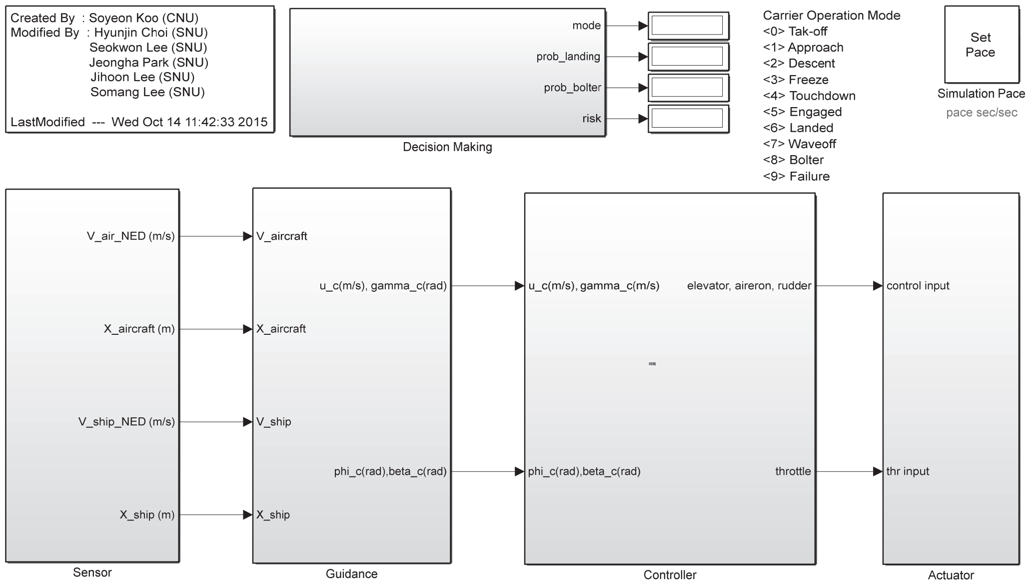

Fig. 4(a): The architecture of the simulator for aircraft carrier air operations established in Simulink®.

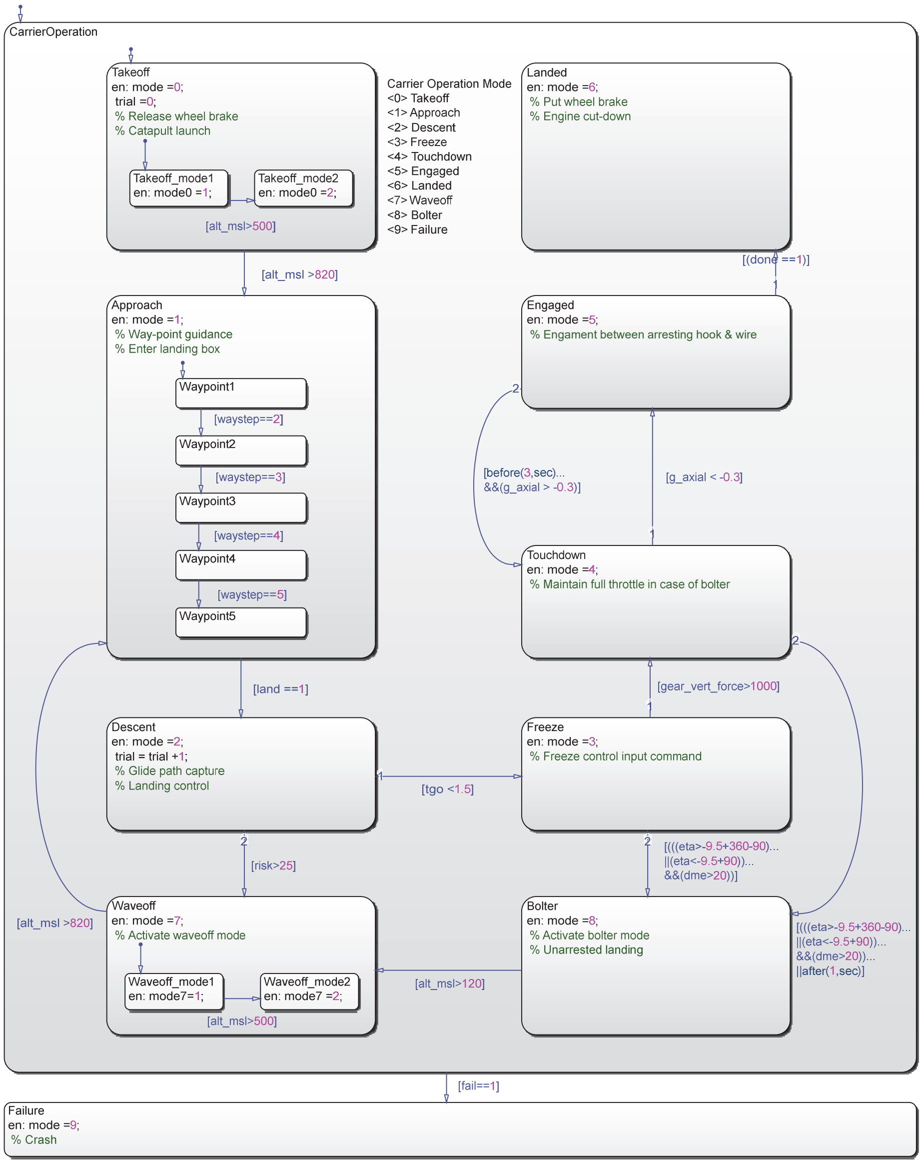

Fig. 4(b): The decision making system modelled in Stateflow®.

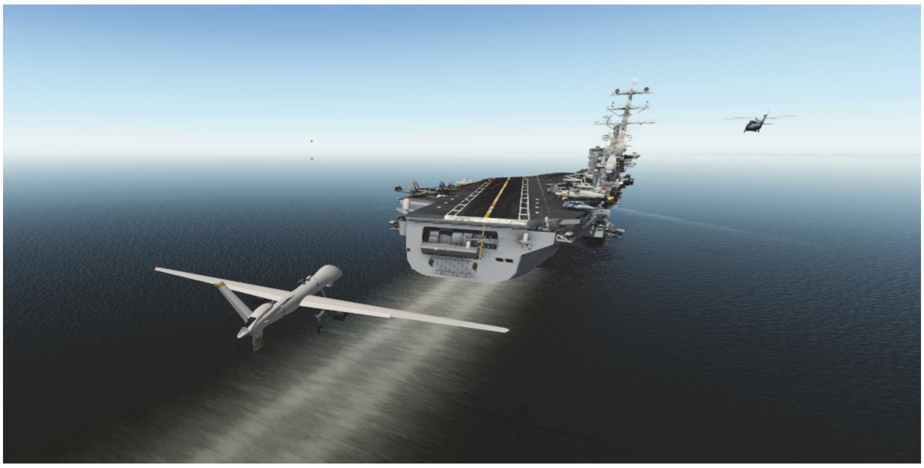

Fig. 4(c): A fixed-wing UAV (MQ-9 Reaper) approaching an aircraft carrier (USS Nimitz) in X-Plane flight simulator.

Table 3: The comparison table of Beaufort force, Pierson-Moskowitz sea state, and X-Plane turbulence level

Beaufort force	Sea state	Turbulence level (X-Plane)	Wind speed (knot, m/s)	
0	0	0.0 - 0.1	< 0.6	< 0.3
1	0	0.1 – 0.9	0.6 - 3	0.3 – 1.5
2	1	0.9 – 2.0	3 - 6.4	1.5 – 3.3
3	2	2.0 – 3.1	6.4 - 10.6	3.3 – 5.5
4	3	3.1 – 4.8	10.6 - 15.5	5.5 – 8
5	4	4.8 – 6.6	15.5 - 21	8 – 10.8

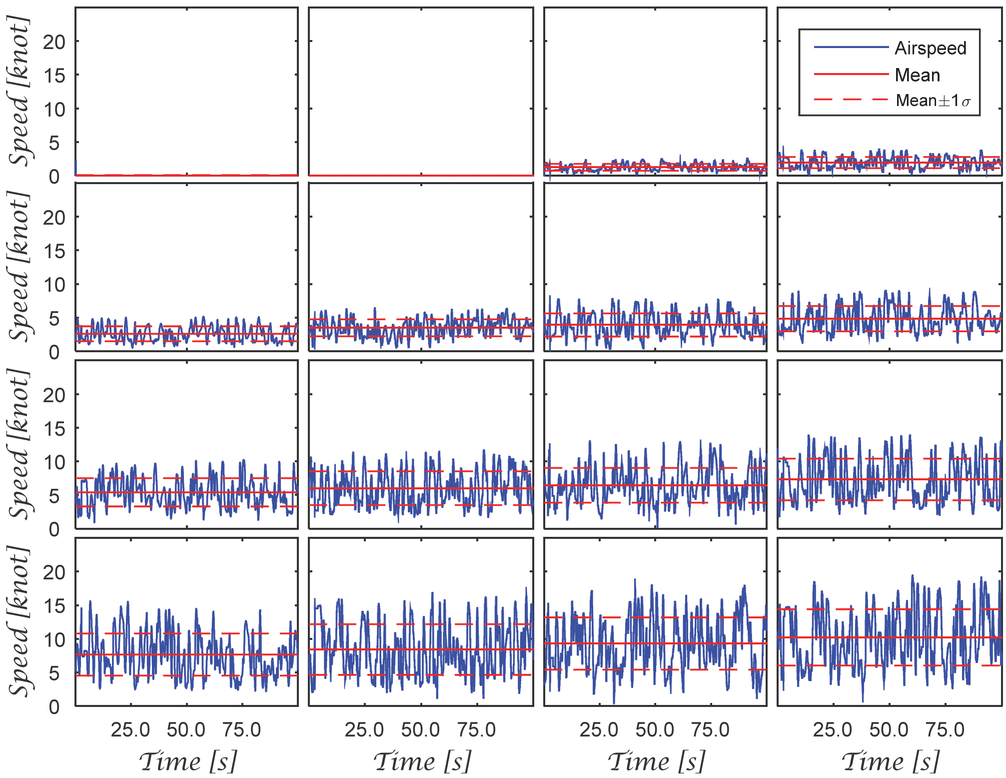

Fig. 5: The turbulence model of X-Plane

The probabilities evaluated by the proposed risk evaluation scheme are in good agreement with those obtained from Monte Carlo simulations as shown in Fig. 6, which has 3 columns and 16 rows of graphs. Each column corresponds to the probability of land, bolter, and fail cases, and each row corresponds to the specific turbulence level as written on the right side of the figure, respectively. Blue dash-dot lines represent the actual occurrence ratio, which are computed from the total number of the occurrence of each case divided by the total number of the landing attempts at that point. Red solid lines represent batch-averaged value of estimated probability; thus, red lines should coincide with blue dash-dot lines if the risk is precisely evaluated for all cases.

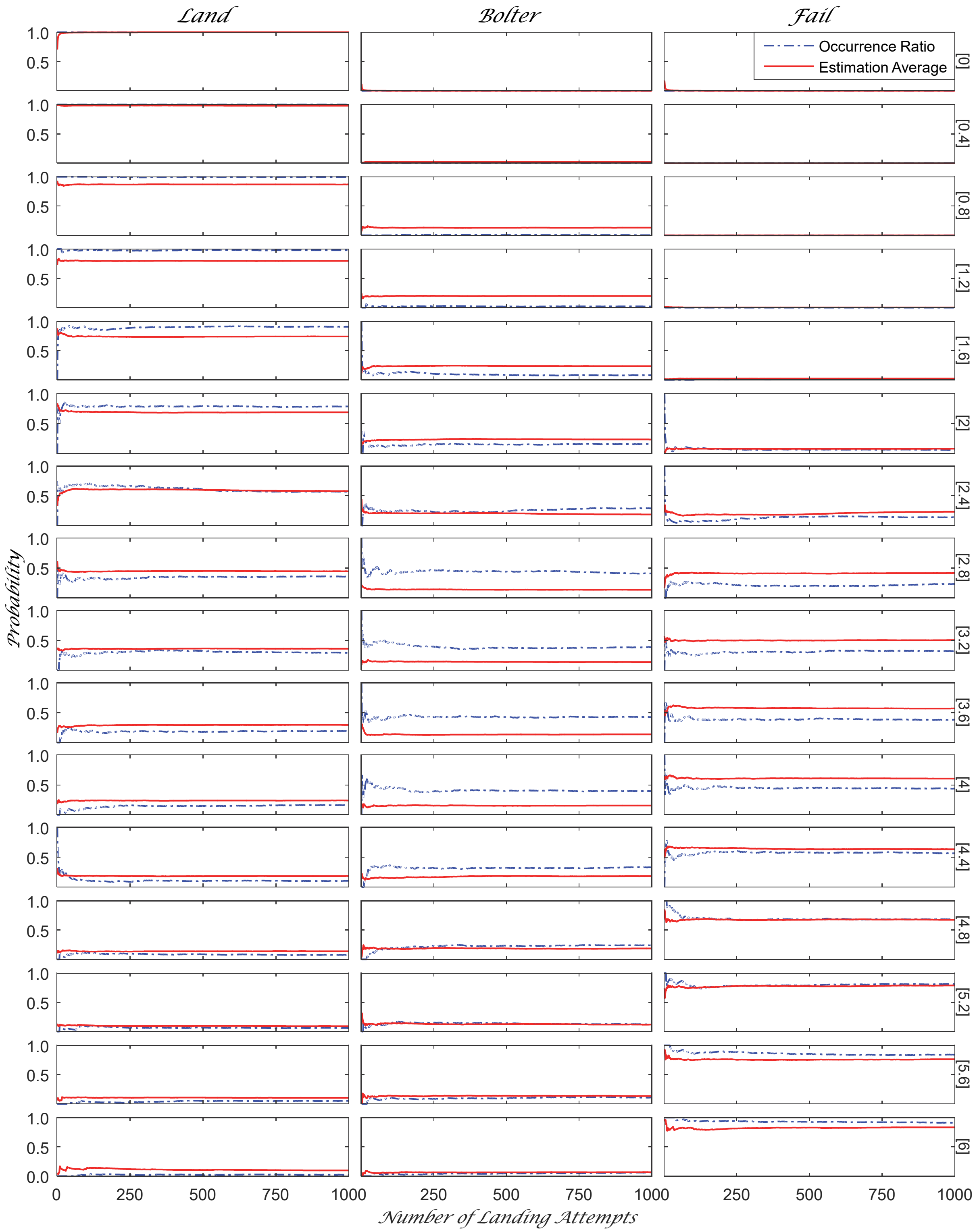

Fig. 6: Comparison of estimated probability with actual occurrence rate.

The final values of all plots in Fig. 6 are extracted and plotted in Fig. 7, where the general tendency of the estimated probabilities and actual occurrence ratio could be seen. There is a certain positive correlation between them, but it is also obvious that the correlation is not perfect and shows minor but not negligible discrepancies.

By comparing the three estimated probabilities and choosing the maximum value, a prediction for the consequence of current landing attempts can be made. The accuracy of this prediction can be estimated by examining Fig. 7, which is shown in Fig. 8. The interesting point in this data is that the prediction tends to be inaccurate near turbulence level 3.0, but becomes accurate as moving to both ends. It should, however, be noted that there is a theoretical upper/lower bounds of the prediction accuracy, which is the maximum/minimum value among the three probabilities. If the results is predicted in a completely random manner without any information at all, the accuracy will be the same as the lower bound. Likewise, even though one knows exactly what kind of result is most probable, the expectation inevitably has a probability to be wrong as much as that probability subtracted from unity.

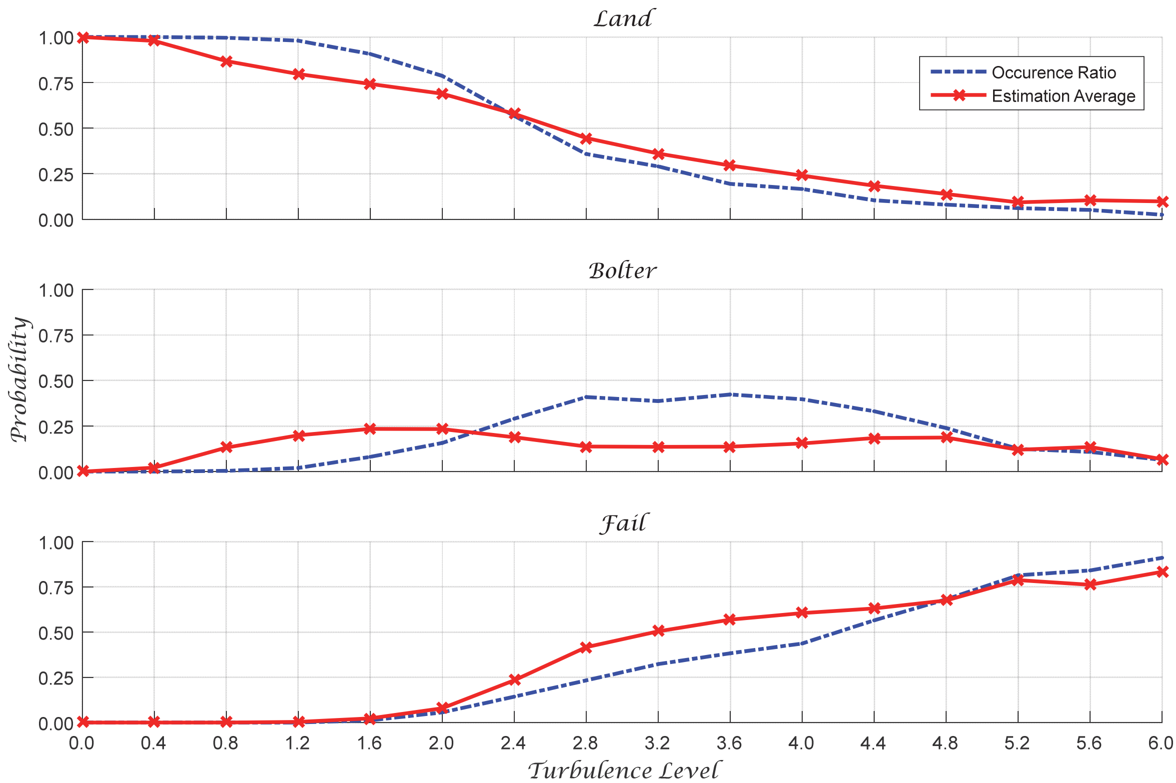

Fig. 7: The tendencies of estimated probability and actual occurrence rate.

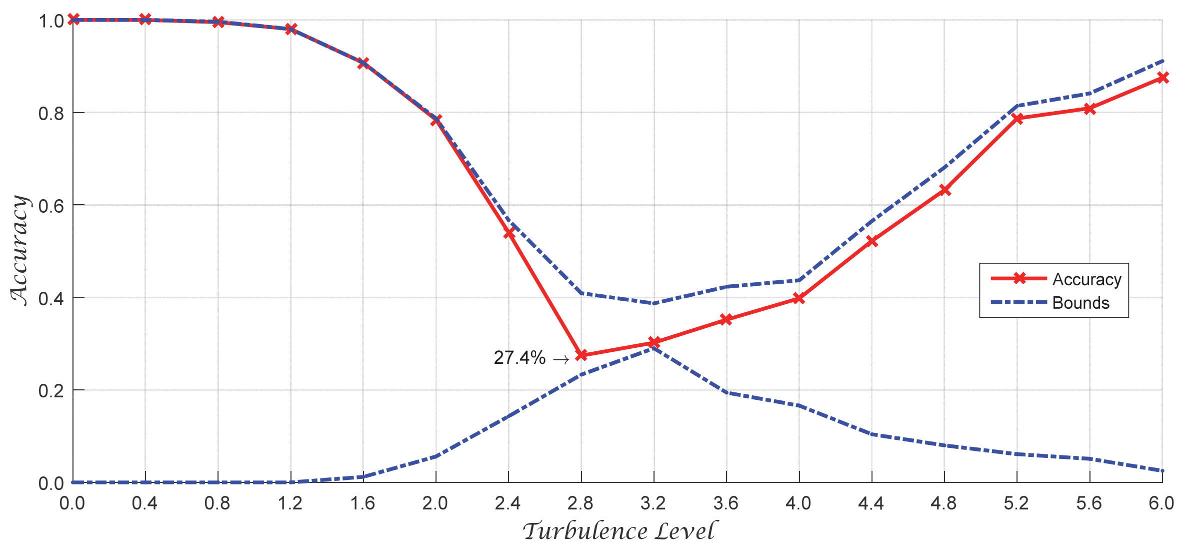

Fig. 8: The *accuracy of the result prediction.*

The performance of the proposed risk evaluation scheme is affected by design parameters and type of a probability density function. The design parameters of the risk evaluation scheme are $\phi_{lo}, \phi_{up}, \theta_{lo}, \theta_{up}, v_{s,lo}, v_{s,up}$, $\varepsilon_{v,lo}, \varepsilon_{v,up}, \varepsilon_{h,lo}, \varepsilon_{h,up}$, and $\bar{\dot{h}}_{lo}$. Although most of them can be determined directly from the dimensions of the target aircraft and the target aircraft carrier, one can adjust the parameters considering the purpose and circumstances. This characteristics enables several possible explanations for the discrepancies in Fig. 7. In the simulations, there may exist more factors influencing the result of the landing attempts such as bouncing of the aircraft, which cannot be addressed by the parameters used in this study.

Conclusion

In this study, the key variables were defined, which variables embrace almost every aspect of possible failures. It was assumed that these variables are normally distributed so that the probabilities of the three distinct outcomes can be estimated with little computation load. Then the wave-off decision strategies using the computed probabilities were proposed. In the simulations, UAV did not actually enter into a wave-off pattern even in the worst case. Therefore, the situation finally falls into one of the three cases so that it can be found out what the outcome actually is. The simulation results show that there is a reasonable agreement between a prediction and an outcome, which proves the validity of the proposed risk evaluation scheme. The deck motion compensation is generally added to the glide path at the final stage of aircraft carrier landing, within approximately 12.5 seconds before touchdown. Therefore, there might be a discrepancy between the tendency of the outcome and prediction if the deck motion is not small enough. In order to consider ship motions, the risk evaluation scheme should explicitly incorporate the guidance law, which determines how to react against and compensate the carrier motions.

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